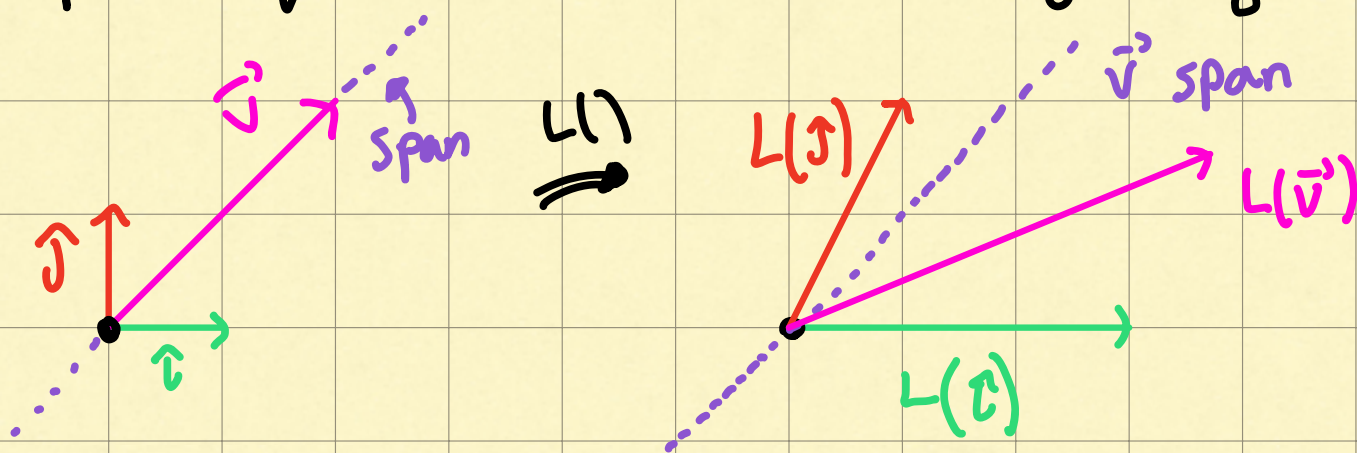


*14 Eigen vectors

Pre requisites: matrices as linear transformations, determinant, linear systems, Change of basis.

consider $L() = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ bc $L(\hat{i}) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ & $L(\hat{j}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

span of vector = line through origin & tip



• most vectors get knocked off span after linear transformation

• what if where $L(\vec{v})$ lands was on $\text{span}(\vec{v})$?

- some special vectors do remain on own span after linear transformation

- $\therefore$ matrix effect on vector = stretch or squish like a scalar.
You stayed on the span

- in this example:

$\text{span}(\hat{i}) = x \text{ axis}$, $L(\hat{i}) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, still in $\text{span}(\hat{i})$

- any other vector on x axis also just stretched by factor of 3..

also in this example:

$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$, $L(\vec{w})$ on span of $\vec{w}$

b/c linearity,
any vector on $\text{span}(\vec{w})$
also stays on

• for this example transformation, those are the only vectors w/ this property of staying on their span:

- i) those on x axis, stretched by 3
- ii) those on $y = -x$, stretched by 2

any other vector rotated during transformation, knocked off line it spans

special vectors are eigenvectors of the transformation, eigenvalue = factor by which its scaled by during transformation

- You can have negative eigenvalues.
ex: $-1/2 \Rightarrow L(\vec{v})$ Flipped & squished,
but stays on $\text{span}(\vec{v})$.

- why this is useful. consider 3D rotation.

if you can find an eigenvector for this rotation, a line that stays on its span...

you have found the axis of rotation !!!

easier to think about $\rightarrow$ than saying
"rotate 30 around $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ $\begin{bmatrix} : \\ : \\ : \end{bmatrix}$

- rotations never stretch or squish anything

- w/ any $L()$, you can define w/ where your basis vectors land...

but to define w/out emphasizing coords

find eigenvectors & eigenvalues...

transformation
matrix

eigen
value

$$A \vec{v} = \lambda \vec{v}$$

eigen
vector

matrix vector
product

scaling eigenvector
by eigenvalue.

$$A \vec{v} = \lambda \vec{v}$$

Scalar
multiple

• finding λ & $\vec{v}$ of A , $\Rightarrow$ what makes
this true

• ask bc mat. vec. mul. on left,
scalar. mul on right

- Swap right using matrix which has effect of scaling any vector by factor of lambda -

col of that matrix : what happens to each basis vector.

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \rightarrow \lambda I. \quad \text{so} \quad \begin{aligned} A\vec{v}' &= \lambda\vec{v}' \\ A\vec{v}' &= (\lambda I)\vec{v}' \end{aligned}$$

now rewrite: $A\vec{v}' - (\lambda I)\vec{v}' = \vec{0}$

$$\underbrace{(A - \lambda I)}_{\text{new matrix}} \vec{v}' = \vec{0}$$

looking for $\vec{v}'$ s.t. new matrix $\cdot \vec{v}' = \vec{0}$.

we want a non zero eigen vector

- we want to become zero.—
- only for product of matrix & non zero vector to become a zero... is if

• transformation associated w/ that matrix to squash into lower dimension $\rightarrow \det(\underbrace{A - \lambda I}_{\text{new matrix}}) = 0.$

- we want tweaked transformation to crush space into lower dimension
- if value exists, $\vec{v}$ is eigen vector of A , staying on own space during transformation A

- there could be no eigenvectors.

ex: rotation by 90° , takes every vector off of its own plane

- meanwhile $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow$ x axis unaffected, all vectors on x axis unchanged, all eigen vectors

- a single eigen value can have more than a line of eigen vectors

ex: $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ "scale everything by 2"

→ every vectors is eigen vector.

→ but only eigenvalue is 2

- eigenbasis

- what if basis vectors are eigenvectors

ex $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$, both scalar multiples,

diagonals are the eigenvalues

in a diagonal matrix

- Diagonal matrices nice to compute high powers of

- rare for basis vectors to be eigenvectors---

- So do a change of basis!

- if your transformation matrix has enough eigenvectors, that span the full space (so you can still reach anything), change coordinate system s.t. eigen vectors are basis vectors

How to express translation in another system:

→ coords of vector to be as new basis

→ in this case our two eigen vectors

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ & $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, make cols of matrix

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} \quad \quad \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

inverse change of basis matrix	original transform	change of basis matrix
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= $\begin{bmatrix} \quad \quad \end{bmatrix}$ same transform,
from perspective
of new basis
vector coordinate system

• when you do this w/ eigenvectors,
new matrix is guaranteed
diagonal w/ eigenvalues
down diagonal. bc
it represents working
in a coordinate system
where basis vectors
are scaled during
transformations.

eigenbasis = basis vectors re eigen
vectors.

now if you need A^{100} , you'll do

$$A \xrightarrow[\text{eigen}]{\text{c.o.b}} A^{100} \xrightarrow{\text{c.o.b}} A^{100}$$

convert basis to make my

- Some transformation you can't do this. ex a shear transform doesn't have enough eigenvectors for a change of basis to eigen basis.